A KEYSTONE INTERACTION IN THE ASSEMBLY OF THE TYPE VI SECRETION SYSTEM MEMBRANE COMPLEX IN *B. FRAGILIS*

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Keywords: Type VI secretion system; Bacteroidetes; membrane complex; protein-protein interaction; VWA domain

ABSTRACT

The Type VI Secretion System (T6SS) is a bacterial nanomachine essential for interbacterial competition through secretion of antagonistic effectors into the target. This nanomachine requires a specialized membrane complex (MC) to anchor the machinery to the cell envelope, and allow the passage of the “poisoned harpoon” (ie. The effector loaded Hcp tube) through the double membrane. While Proteobacteria possess a characterized MC (TssJLM), Bacteroidetes encode a distinct, non-homologous MC (TssNQOPR) whose assembly and functional architecture remain poorly studied. In our latest work, we investigated core protein-protein interactions within the Bacteroidetes MC: the TssR von Willebrand Factor A (VWA) domain and the TssP C-terminal domain (CTD). We characterized this interaction through co-purification, isothermal titration calorimetry, and native mass spectrometry. We validated the Alphafold structural predictions through directed mutagenesis, demonstrating that mutations at the predicted interface abolish binding. Finally, we explored the evolutionary intricacies of these domains and their phylogeny to understand how they emerged and evolved. In this study, we showcased the first direct biochemical evidence of an interaction between two essential T6SS MC components in Bacteroides fragilis.

REFERENCES

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